INCH-POUND
MIL-DTL-24643/47C
22 August 2002
SUPERSEDING
MIL-C-24643/47B
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1SMWU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First -	Copper conductor, tin coated, ASTM B 286, AWG Size 22, 7 strands.
Second -	Separator may be used at manufacture's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation. Nominal wall thickness 0.030 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements.
Fourth -	Braided shield of AWG No. 34 or No. 36 tin coated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
Fifth -	Shield insulation of two polyester tapes, per NEMA FI 4, outer tape sealed. An alternate shield insulation may consist of one polyester tape, per NEMA FI 4, plus extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

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Construction (watertight) (continued)

Sixth - The required number of singles (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with No. 1, from the center outward. Fillers shall be used as necessary to make a firm, well rounded assembly.

Seventh - Binder tape, applied helically with overlap.

Eighth - Cross-linked polyolefin jacket. Jacket thickness, 0.50 inch minimum average.

TABLE I. Details.

Military part no. M24643/47	Size	Number of singles	Overall diameter (inches)	
			TYPE LS1SMWU	
			min	max
-01UN	70	70	1.465	1.555

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor resistance - ohms/1000 feet at 25°C, maximum	17.71
Voltage withstand - volts, root mean square, minimum	
Conductor to shield	5000
Shield to shield	500
Insulation resistance - Megohms -1000 feet minimum.	
Conductor to shield	500
Shield to shield	100
Conductor and shield continuity	No failure

Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-DTL-24643 for limits of water leakage	No failure
Capacitance - 1 kilohertz (kHz), pF/ft, maximum	35
Attenuation - at 400 MHz, decibel per 100 ft, maximum	18

Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Cold bending, cable - at minus 0 ± 2°C,	
mandrel maximum diameter 20 inches	No damage
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35

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Physicals (unaged)
- Insulation
  - Tensile strength - lb/in^2, minimum 1800
  - Elongation - percent, minimum 250
- Jacket (cable)
  - Tensile strength - lb/in^2, minimum 1300
  - Elongation - percent, minimum 160

Group C:

Physicals (aged)
- Insulation
  - Air oven
    - Tensile strength - percent of unaged, minimum 80
    - Elongation - percent of unaged minimum 80
- Jacket (cable)
  - Air oven
    - Tensile strength - percent of unaged, minimum 60
    - Elongation - percent of unaged minimum 60
  - Hot oil immersion
    - Tensile strength - percent of unaged, minimum 50
    - Elongation - percent of unaged, minimum 50
- Shrinkage ..... No failure
- Heat distortion - percent of unaged, maximum 30
- Permanence of printing (jacket) cycles, minimum 125
- Cable sealant removability ..... No failure
- Shield - conformance to material, construction and coverage ..... No failure

Group D:
- Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

- Aging and compatibility (cable) (125 ± 5°C.) ..... No failure
- Abrasion resistance (jacket) - scrapes, minimum 75
- Acid gas equivalent - percent, maximum
  - Jacket 2
  - Fillers 2
  - Insulation 18
  - Shield insulation 2
- Halogen content - percent, maximum
  - Jacket 0.2
  - Fillers 0.2

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Immersion (jacket)
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum 50
Smoke index, maximum
Jacket 25
Fillers 45
Insulation 45
Toxicity index, maximum
Jacket 5
Fillers 5
Insulation 1.5
Durometer (jacket) - (type A) hardness, minimum.....80
Weathering (jacket)..... No failure

UNIT ORDERING LENGTHS: 1000 feet nominal

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-045)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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